

Choosing a Right-Triangle Method

Answer Key

High School Geometry

Quick Answers

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|---------|-------------------|-----------|
| 1. 10 | 5. 22.62 | 9. 7.07 |
| 2. 9.90 | 6. 6 | 10. 53.13 |
| 3. 6.30 | 7. $\frac{8}{15}$ | |
| 4. 12 | 8. 13 | |
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Common wrong answers

3. If they got 5.16: used sine instead of tangent, dividing by the hypotenuse the problem never gave
 4. If they got 17.49: added the squares instead of subtracting, treating the hypotenuse as a leg
 5. If they got 67.38: used inverse cosine on the same ratio, finding the other acute angle
 6. If they got 10.39: used the 30-60-90 ratio the wrong way round and returned the long leg
 8. If they got 10.91: subtracted the squares, treating the wall height as the hypotenuse
 9. If they got 5: halved the diagonal instead of using the 45-45-90 ratio
 10. If they got 36.87: used sine on the half-base over the equal side instead of cosine
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1. Method: Pythagorean theorem.

Step 1. Two legs are given and the hypotenuse is missing, with no angle anywhere in the problem — that is the Pythagorean signature. A trig ratio here would need an angle the problem does not supply.

Step 2. $c^2 = a^2 + b^2 = 6^2 + 8^2 = 36 + 64 = 100$.

Step 3. $c = \sqrt{100} = \boxed{10}$ (the 3–4–5 triple scaled by two, so the whole-number answer is a good sign).

2. Method: special triangle (45–45–90).

Step 1. The right angle sits at C and one acute angle is 45° , so the third angle is also 45° : this is the isosceles right triangle, whose sides are in the ratio $1 : 1 : \sqrt{2}$.

Step 2. In that ratio the hypotenuse is $\sqrt{2}$ times a leg, and the given leg is $b = 7$, so $c = 7\sqrt{2}$.

Step 3. $c = 7\sqrt{2} \approx \boxed{9.90}$

3. Method: trig ratio (tangent).

Step 1. An angle of 35° is given — not a special angle — with the side *adjacent* to it, and the side *opposite* it is wanted. The ratio linking opposite to adjacent is the tangent, so no hypotenuse is needed.

Step 2. $\tan 35^\circ = \frac{a}{b} = \frac{a}{9}$, so $a = 9 \tan 35^\circ$.

Step 3. $a \approx \boxed{6.30}$

If they got 5.16: used sine instead of tangent — sine would need the hypotenuse, which this problem never gives.

4. Method: Pythagorean theorem (missing leg).

Step 1. Again two sides and no angle, so Pythagoras — but here one of the given sides is the *hypotenuse* c , so the theorem is rearranged rather than applied straight.

Step 2. $b^2 = c^2 - a^2 = 15^2 - 9^2 = 225 - 81 = 144$.

Step 3. $b = \sqrt{144} = \boxed{12}$

If they got 17.49: added the squares instead of subtracting, which treats the hypotenuse as a leg.

5. Method: inverse trig ratio.

Step 1. Here the missing quantity is an *angle* and two sides are known, so the tool is an inverse ratio. Relative to angle A , the side $a = 5$ is opposite and $c = 13$ is the hypotenuse — opposite over hypotenuse is sine, so use $\sin^{-1}$.

Step 2. $\sin A = \frac{5}{13}$, so $A = \sin^{-1}\left(\frac{5}{13}\right)$.

Step 3. $A \approx \boxed{22.62^\circ}$

If they got 67.38° : applied $\cos^{-1}$ to the same ratio, which returns the other acute angle B .

6. Method: special triangle (30–60–90).

Step 1. A 30° acute angle with the right angle at C makes this the half-equilateral triangle, sides in the ratio $1 : \sqrt{3} : 2$ — exact, so no calculator is needed.

Step 2. The side opposite the 30° angle is the short leg, and it is half the hypotenuse. Here a is opposite $A = 30^\circ$ and $c = 12$.

Step 3. $a = \frac{1}{2}(12) = \boxed{6}$

If they got 10.39: used the ratio the wrong way round and returned the long leg $b = 6\sqrt{3}$.

7. Method: definition of the ratio (no computation).

Step 1. The question asks for a ratio, not a length, so nothing needs solving — read the two sides off the figure relative to angle A .

Step 2. Opposite A is $a = 8$; adjacent to A (and not the hypotenuse) is $b = 15$. Tangent is opposite over adjacent.

Step 3. $\tan A = \boxed{\frac{8}{15}}$ (already in lowest terms, since 8 and 15 share no factor).

8. Method: Pythagorean theorem in context.

Step 1. Wall, ground and ladder form a right triangle; the wall is vertical so the right angle is between wall and ground. Two legs are known and the ladder is the hypotenuse, and no angle is given — Pythagoras.

Step 2. $\text{ladder}^2 = 5^2 + 12^2 = 25 + 144 = 169$.

Step 3. $\text{ladder} = \sqrt{169} = \boxed{13 \text{ ft}}$

If they got 10.91: subtracted the squares, which treats the 12 ft wall height as the hypotenuse — but the ladder is the longest side.

9. Method: special triangle (45–45–90) in context.

Step 1. A square's diagonal makes a 45° angle with each side, so half the square is the isosceles right triangle: the diagonal is the hypotenuse and the two sides are the equal legs.

Step 2. With hypotenuse 10, a leg is $10 \sin 45^\circ = \frac{10}{\sqrt{2}} = 5\sqrt{2}$.

Step 3. side $= 5\sqrt{2} \approx \boxed{7.07 \text{ cm}}$

If they got 5: halved the diagonal, which is what a rectangle's other diagonal would do, not the 45–45–90 ratio.

10. Challenge — method: split the isosceles triangle, then cosine.

Step 1. The triangle is not right angled, so no single ratio applies to it as drawn. The altitude from C to the base c bisects the base (because the two slanted sides are equal), producing two congruent right triangles — and *that* is where a ratio can be used.

Step 2. In one of those right triangles, the base angle A sits with an adjacent side of $\frac{1}{2}(12) = 6$ and hypotenuse 10. Adjacent over hypotenuse is cosine, so $\cos A = \frac{6}{10}$.

Step 3. $A = \cos^{-1}(0.6) \approx \boxed{53.13^\circ}$, and by symmetry the other base angle is the same.

If they got 36.87° : used $\sin^{-1}$ on the same ratio, which gives the angle at the apex half, not the base angle.